

TUGAS K32 FISIKA

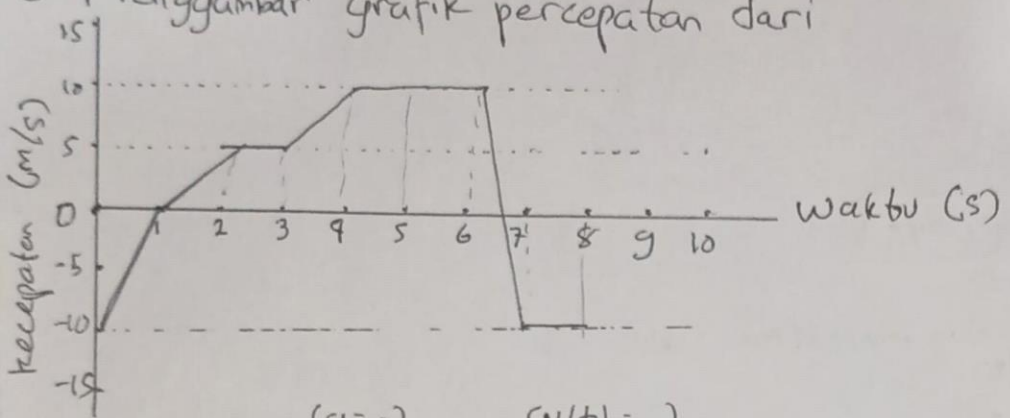
TUGAS 01

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Fly Samudra

Jawaban

3. Menggambar grafik percepatan dari



Menghitung gradien dan persamaan saat, maka grafiknya

a. $t = 0-1$ $a = 10$

0-1	10
1-2	5
2-3	0
3-4	5
4-5	0
5-6	0
6-7	-20
7-8	0

($v(t) = \dots$)

$$v(t) = -10 + 10t$$

$$= -5 + 5t$$

$$= 5$$

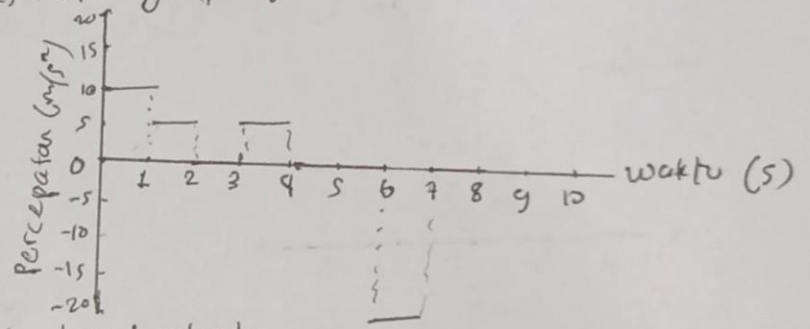
$$= -10 + 5t$$

$$= 10$$

$$= 10$$

$$= 10 - 20t$$

$$= -10$$



b. Benta saat $t = 8s$ bisa menggunakan integral; jika $x(0) = 2m$

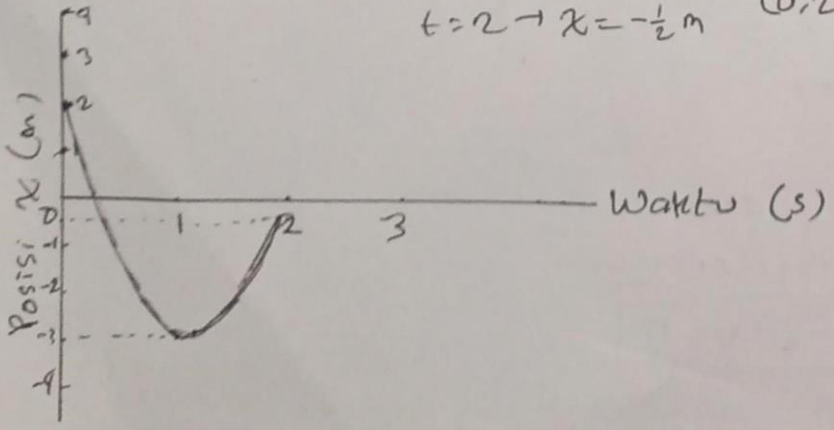
$$x(t) = 2 + \int_0^t v(t) dt$$

$$= 2 + \left(-\frac{10}{2} + \frac{5}{2} + 5 + \left(5 + \frac{5}{2} \right) + 10 + 10 + 0 - 10 \right)$$

$$= 2 + 20$$

$$= \boxed{22 \text{ m}}$$

c. Hitung posisi saat $t=0 \rightarrow x=2m$ didapat titik-titik
 $t=1 \rightarrow x=-3m$ $(0, 2); (1, -3); \text{ dan } (2, -\frac{1}{2})$
 $t=2 \rightarrow x=-\frac{1}{2}m$



$$7. \vec{r} = 3t\hat{i} + 4t^2\hat{j} + 2\hat{k}$$

a. Menentukan $\vec{v}(t)$

$$\begin{aligned}\vec{v}(t) &= \frac{d\vec{r}(t)}{dt} = \frac{d}{dt} (3t\hat{i} + 4t^2\hat{j} + 2\hat{k}) \\ &= 3\hat{i} + 8t\hat{j} + 0\hat{k} \\ &= \underline{3\hat{i} + 8t\hat{j}} \text{ m/s}\end{aligned}$$

b. $\vec{v}$ dan $\|\vec{v}\|$ saat $t=2$

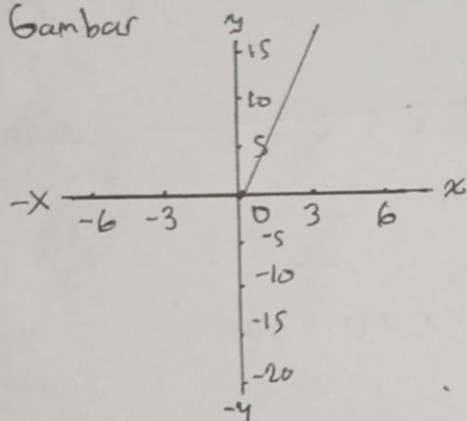
$$\begin{aligned}\vec{v}(t)|_{t=2} &= \vec{v}(2) \\ &= 3\hat{i} + 8(2)\hat{j} \\ &= \underline{3\hat{i} + 16\hat{j}} \text{ m/s}\end{aligned}$$

$$\begin{aligned}\|\vec{v}(2)\| &= \sqrt{3^2 + 16^2} \\ &= \sqrt{9 + 256} \\ &= \sqrt{265} \approx \underline{16,27882 \text{ m/s}}\end{aligned}$$

c. Menggambar vektor dan menentukan sudut

$$\begin{aligned}\arg(\vec{v}(2)) &= \tan^{-1}\left(\frac{\vec{v}_y(2)}{\vec{v}_x(2)}\right) \\ &= \tan^{-1}\left(\frac{16}{3}\right) \approx \underline{79,3803^\circ}\end{aligned}$$

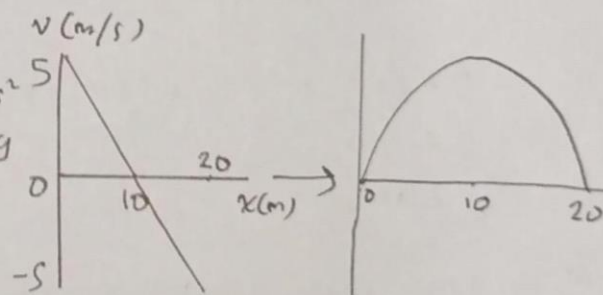
Gambar



(Referensi pakai soal kuis sehingga awal $g=10 \text{ m/s}^2$)

10. $O(0,0)$ dan v_0, θ_0 dengan grafik di samping

$$\begin{aligned}a. v_{0y} &= v_{0y}(0) = \underline{5 \text{ m/s}} & -a &= a - gt \\ v_{0x} &= v_x = \frac{20}{t} = \underline{20 \text{ m/s}} & -s &= s - 10t \\ & & t &= 1 \text{ m/s}\end{aligned}$$



b. Sudut kemiringan awal $= \theta_0$

$$\begin{aligned}&= \tan^{-1}\left(\frac{v_{0y}}{v_{0x}}\right) \\ &= \tan^{-1}\left(\frac{5}{20}\right) = \tan^{-1}\left(\frac{1}{4}\right) \approx \underline{14,0362^\circ}\end{aligned}$$

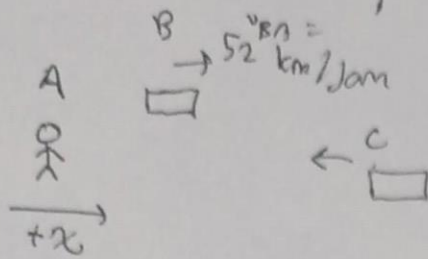
c. Koordinat tertinggi, yaitu saat $x=10$ dan $t=\frac{1}{2}$

$$\begin{aligned}s &= v_0 t - \frac{1}{2} g t^2 \\ &= 5\left(\frac{1}{2}\right) - \frac{1}{2} \cdot 10 \left(\frac{1}{2}\right)^2 \\ &= \frac{5}{2} - \frac{5}{4} \\ &= \frac{5}{4}\end{aligned}$$

Maka, koordinat tertinggi

$$\underline{h_{\max} = (10, \frac{5}{4}) \text{ meter}}$$

13. Soal gerak relatif



a. $V_{BA} = 52 \text{ km/jam}$

$V_{CA} = -78 \text{ km/jam}$

$V_{CB} = V_{CA} - V_{BA}$

$= -78 - 52$

$= \underline{-130 \text{ km/jam}}$

Artinya, kecepatan mobil C menurut B lebih cepat karena berlawanan arah

$\vec{V}_{CB} = (-81,25\hat{i} + 0\hat{j}) \text{ m/s}$

atau $\vec{V}_{CB} = (-130\hat{i} + 0\hat{j}) \text{ km/jam}$

b. $V_{CA} = V_{CA} + at$

$0 = -21,66 + 10 \cdot a$

$\Rightarrow a = +21,66 \text{ m/s}^2$

Ubah dari km/jam ke m/s

$-78 \text{ km/jam} = -21,66 \text{ m/s}$

Maka, $V_{CA}(t) = -21,66 + 21,66t \text{ m/s}$

Untuk $0 \leq t \leq 10$

Untuk menghitung kecepatan rata-rata, integralkan dan bagi dengan Δt

$= \frac{1}{10} \int_0^{10} V_{CA} dt$

$= \frac{1}{10} \left[\frac{21,66t^2}{2} - 21,66t \right]_0^{10}$

$= \frac{1}{10} (-50 \cdot 21,66)$

$= -10,833 \text{ m/s}$

Dengan demikian, kecepatan rata-ratanya adalah

$\vec{V}_{CA} = \underline{(-10,833\hat{i} + 0\hat{j}) \text{ m/s}}$ atau dalam km/jam, $\vec{V}_{CA} = \underline{(-39\hat{i} + 0\hat{j}) \text{ m/s}}$

3. (Soal kuis)

$m_A = 2 \text{ kg}$, $m_B = 8 \text{ kg}$; $\mu_{BS} = 0,5$, $\mu_{BK} = 0,3$

$\mu_{AS} = 0,4$, $\mu_{AK} = 0,2$

Karena ingin bergerak, $a=0$ dan menggunakan gaya gesek statis sehingga

$F - f_B - f_A - T = 0 \Rightarrow F = f_A + f_B + T$

$T = f_A$ karena ingin bergerak sehingga

$F = f_A + f_A + f_B$

$= \mu_{AS} \cdot N_A + \mu_{AS} \cdot N_A + \mu_{BS} \cdot N_B$

$= 2\mu_{AS} \cdot m_A \cdot g + \mu_{BS} \cdot (m_A + m_B)g$

$= 2 \cdot 0,4 \cdot 2 \cdot 10 + 0,5 \cdot 10 \cdot 10$

$= 16 + 50$

$= \underline{66 \text{ N}}$

